Terraform: Getting Started

Introducing IaC and Terraform



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nedinthecloud.com

Provisioning infrastructure through software to achieve consistent and predictable environments.



Core Concepts of IaC

Defined in code

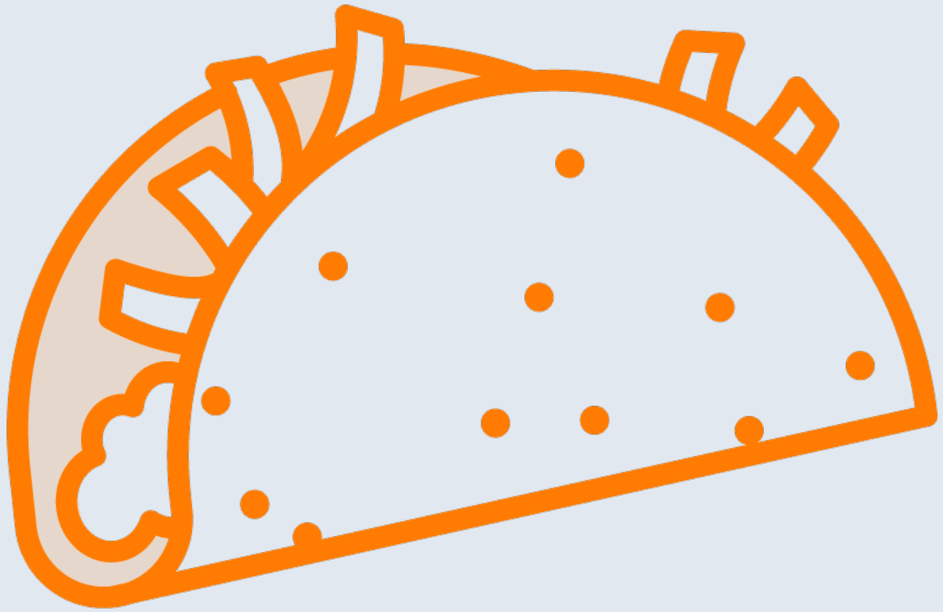
Stored in source control

Declarative vs. imperative

Declarative or Imperative

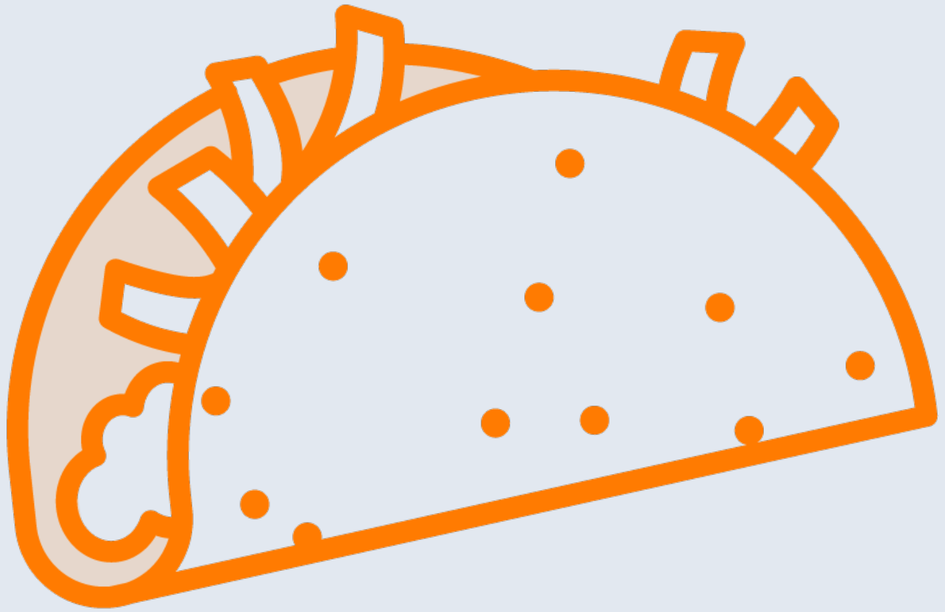
```
# Make me a taco
get shell
get beans
get cheese
get lettuce
get salsa

put beans in shell
put cheese on beans
put lettuce on cheese
put salsa on lettuce
```



Declarative or Imperative

```
# Make me a taco
food "taco" "bean_taco" {
  ingredients = [
    "beans", "cheese",
    "lettuce”, “salsa"
  ]
}
```



Core Concepts of IaC

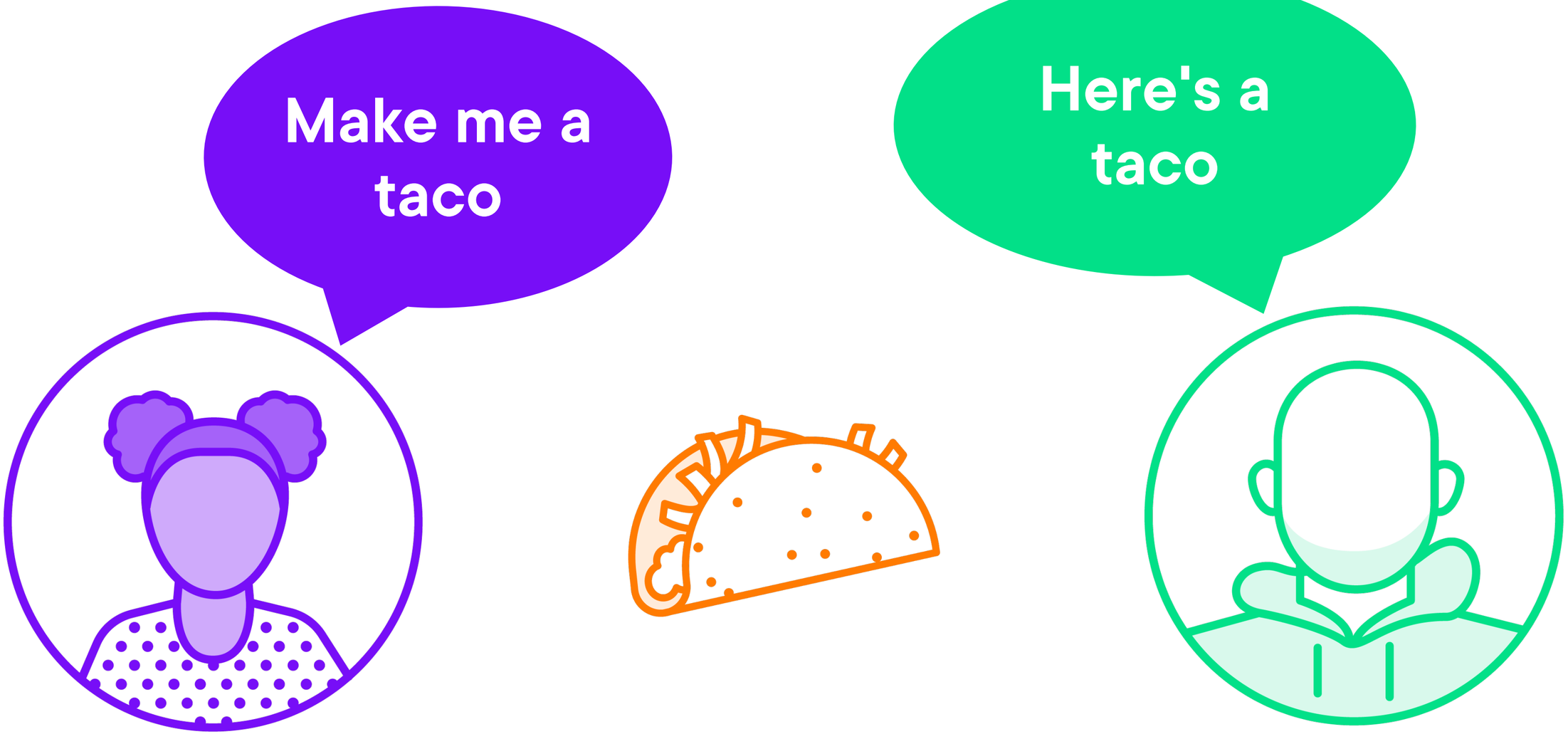
Defined in code

Stored in source control

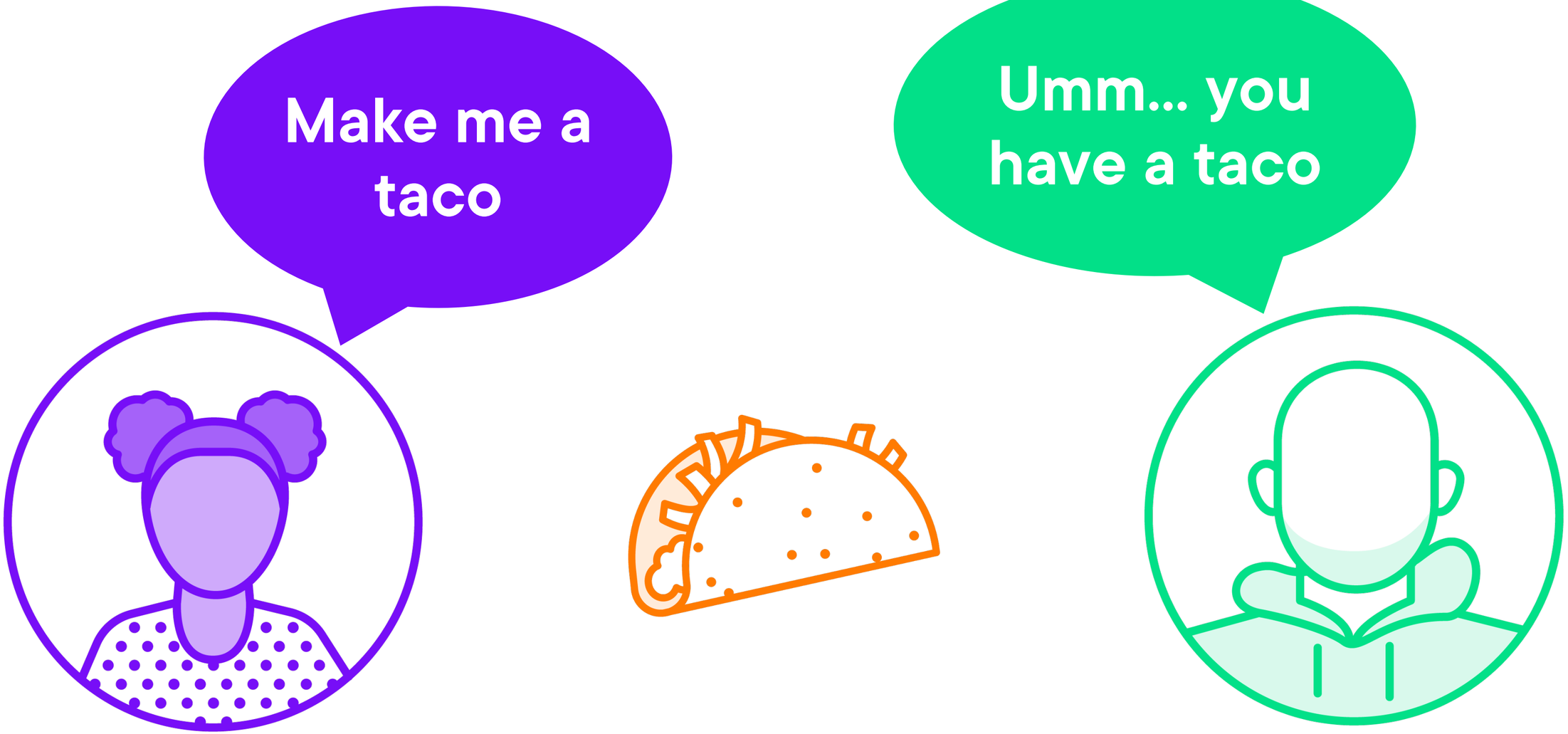
Declarative vs. imperative

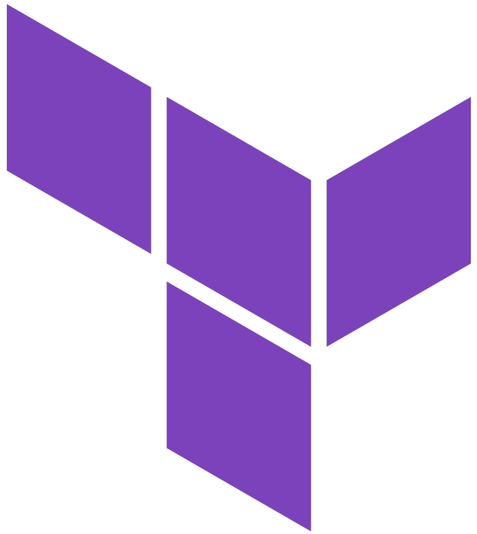
Idempotent and consistent

Idempotent



Idempotent





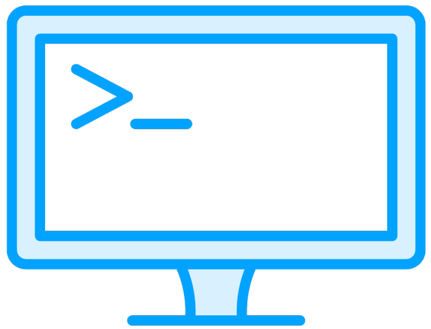
HashiCorp

Terraform

- Infrastructure automation tool
- Terraform Community
- Platform agnostic
- Single binary compiled from Go
- Declarative syntax
- HashiCorp Configuration Language (HCL) or JSON
- No central server or agent

hashicorp.com/brand

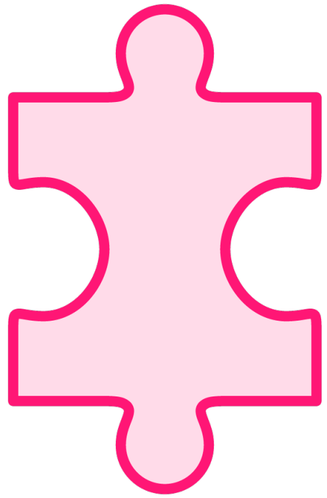
Core Components



Terraform binary



Configuration files



Provider plugins



State data

Terraform Workflow

Write

- Author the code

Plan

- Compare code against state

Apply

- Apply changes and record results



Installation

Download the binary for your platform

Add to your PATH environment variable

Share and enjoy!

Installation and CLI Basics

Learn about installing Terraform and using the basic command line interface.